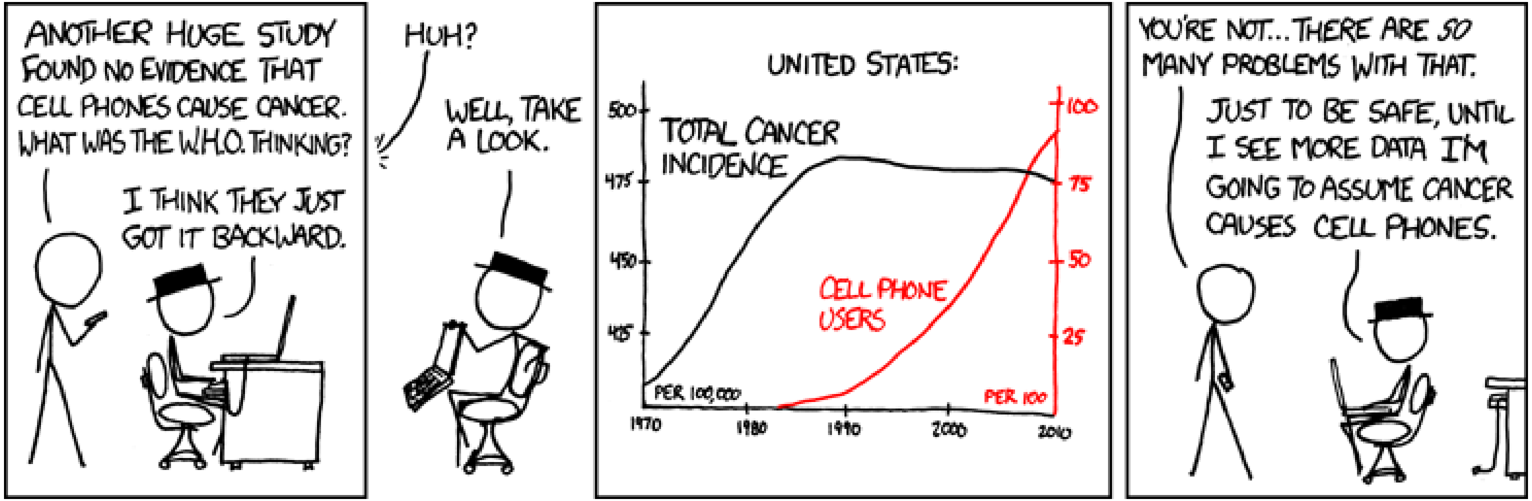
Introduction to Causal Inference for Software Engineering

A 90-minute hands-on tutorial

Julien Siebert · Julian Frattini · Hans-Martin Heyn · Roberto Pietrantuono

Section 3a

Basic Causal Structures



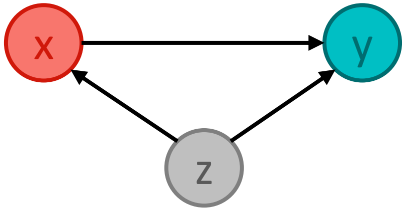
https://xkcd.com/925/

The “building blocks” of causal models

Confounder

(or: fork)

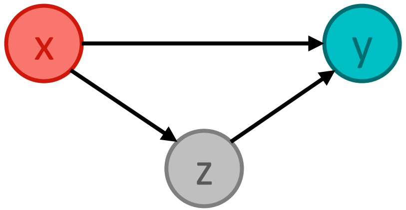
z is a common cause of both exposure x and the outcome y



Mediator

(or: pipe)

z is an effect of exposure x and a cause the outcome y



Collider

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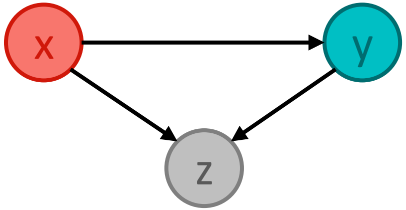
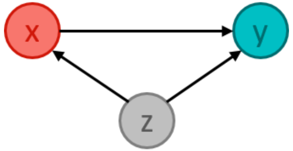
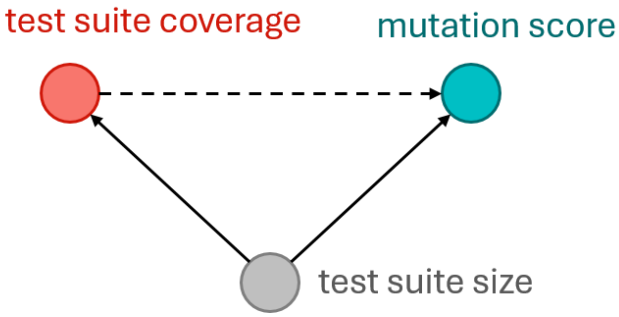




Confounder

- "Coverage Is Not Strongly Correlated with Test Suite Effectiveness"

We found that there is a low to moderate correlation between coverage and effectiveness when the number of test cases in the suite is controlled for. In addition, we found that



Inozemtseva, L., & Holmes, R. (2014, May). Coverage is not strongly correlated with test suite effectiveness. In *Proceedings of the 36th international conference on software engineering* (pp. 435-445).

Most Influential Paper at ICSE'24

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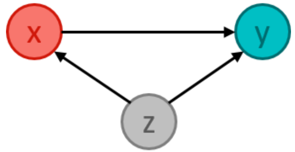
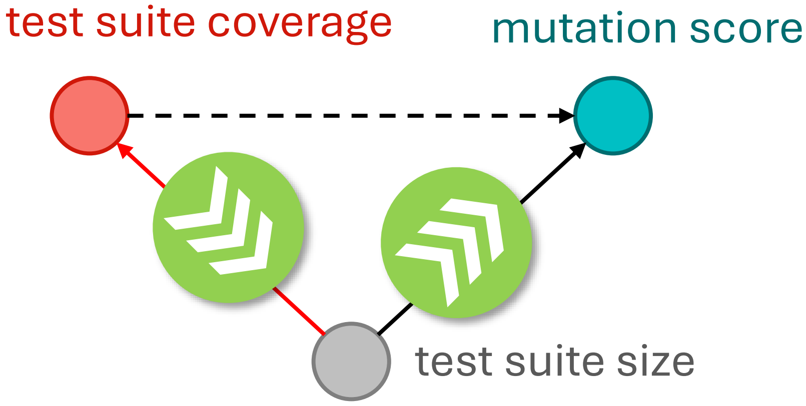
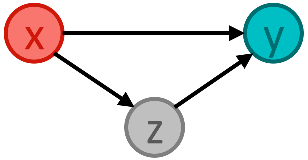


Confounder



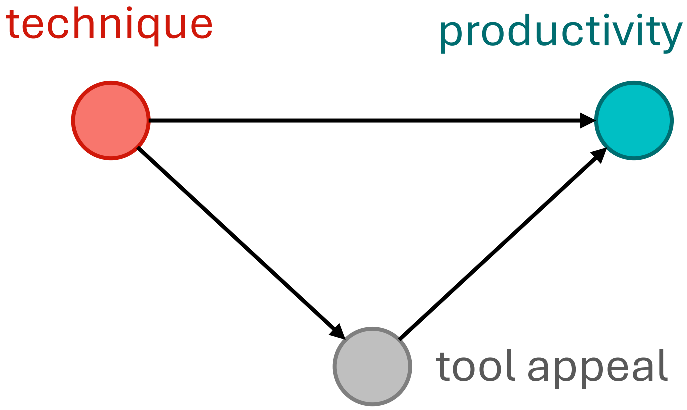
- A higher test suite coverage <u>is associated with</u> test suite size which leads to higher effectiveness (mutation score).
  - **A higher test suite coverage, however, does not cause a higher test suite size!**
- When *not* controlling the confounder, “information flows” from the cause through the confounder to the effect in a non-causal direction.

! If both treatment and outcome have a common cause, we need to statistically adjust for (i.e., control) the confounder to de-bias the effect of interest.



Mediator

- Company Y tested a **new technique** in their requirements engineering phase that is enabled by a **prototype tool**. We investigated whether this technique improved the **engineers' productivity**.”



- On its own, **the new technique lifts productivity.**
- To enforce the technique, a rigid, form-heavy tool is introduced. Engineers dislike the tool.
- A strong tool appeal increases productivity.
- Overall, **the new technique leads to less productivity.**

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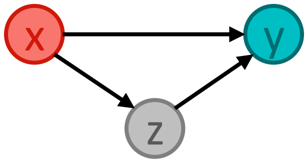
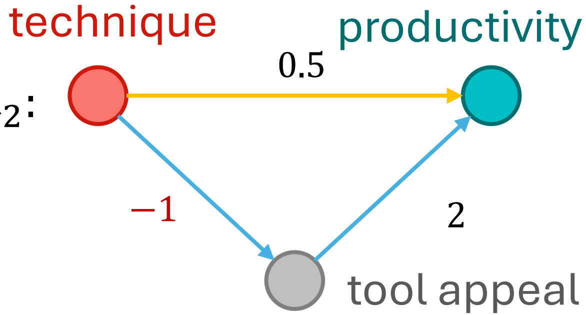


Mediator

m_1:

m_2:



$m_1: productivity \sim technique$

$m_2: productivity \sim technique + appeal$

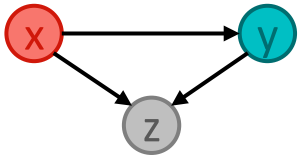
Direct effect (of new technique): 0.5

Indirect effect (mediated by tool appeal): $-1 \cdot 2 = -2$

Total effect = **Direct effect** + **Indirect effect**

$= -1.5 = 0.5 + -2$

! We need to statistically control for a **mediator**, if we want to distinguish between the **direct** and **indirect** effect of a **treatment** on an **outcome**.

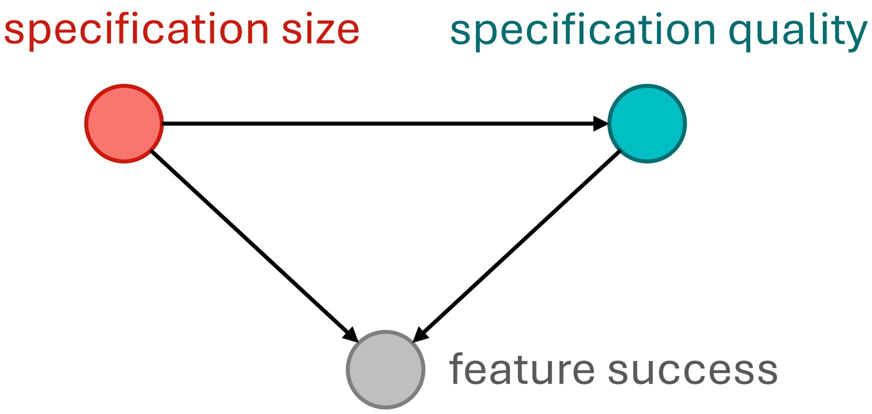


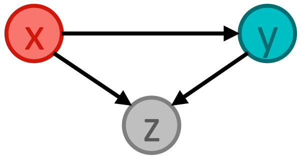
Collider

- Does the **length of a requirements specification** affect its **quality**? Both factors do affect whether the specified feature is implemented successfully.

$m_1: quality \sim size + success$

$m_2: quality \sim size$





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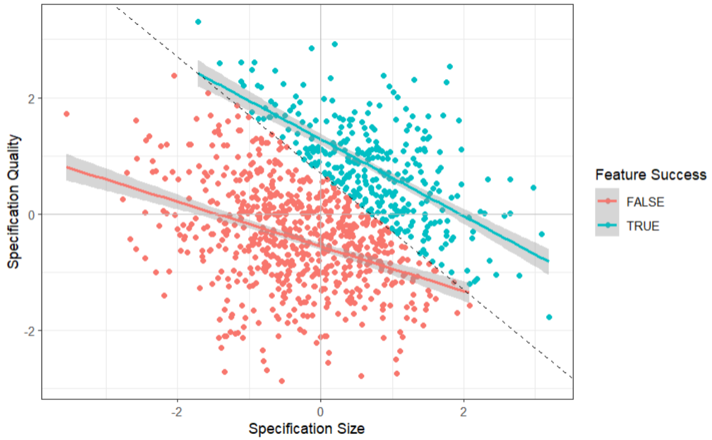
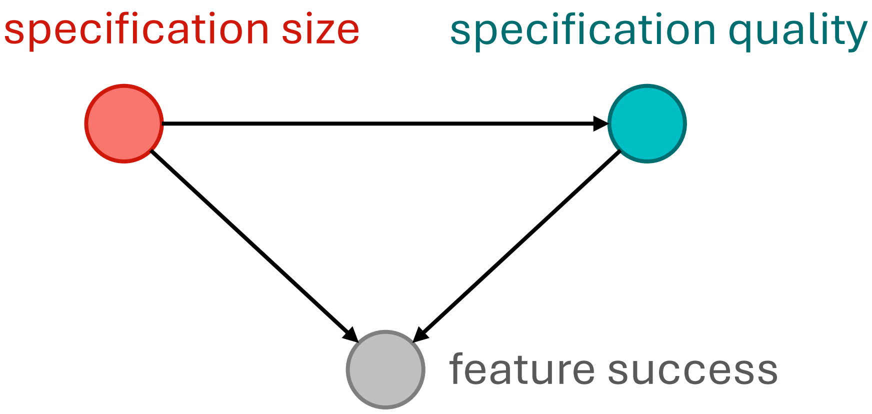
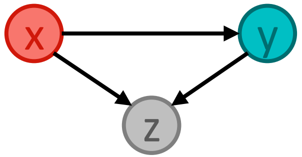


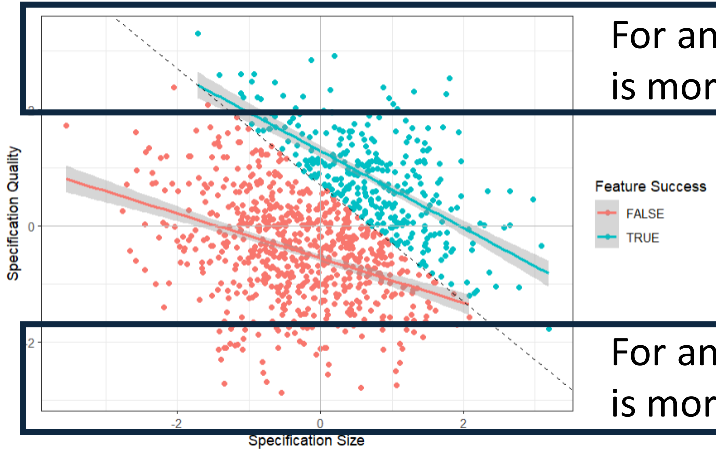




Collider

- Does the **length of a requirements specification** affect its **quality**? Both factors do affect whether the specified feature is implemented successfully.

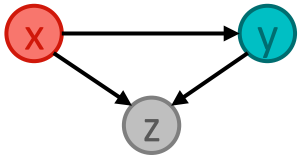
$$m_1 : quality \sim size + success$$



For any high quality value, if feature success is *false*, then it is more likely that the specification size was small.

For any low quality value, if feature success is *true*, then it is more likely that the specification size was large.

Collider

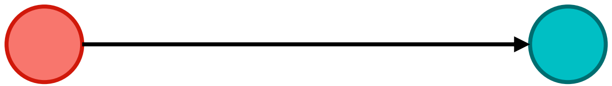


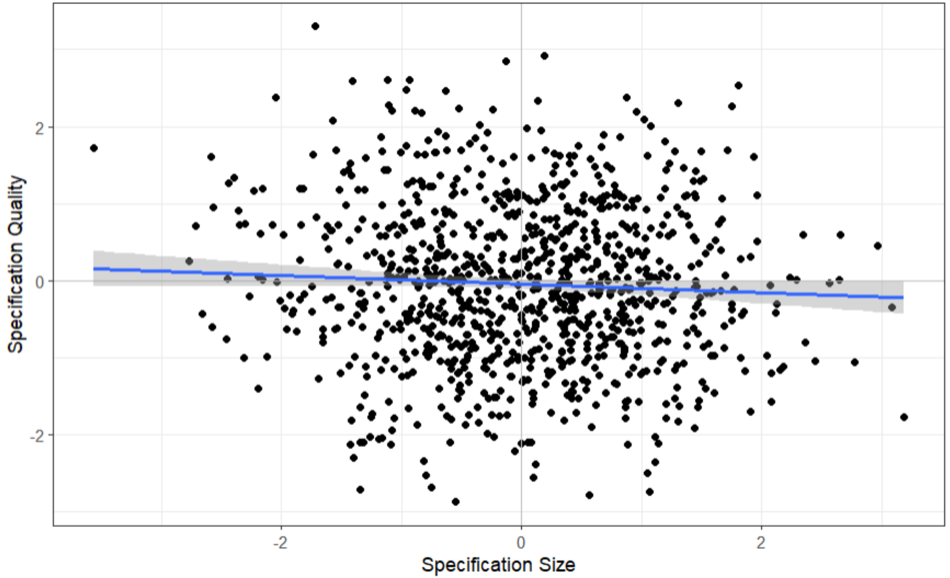
- Does the **length of a requirements specification** affect its **quality**? Both factors do affect whether the specified feature is implemented successfully.

$$m_2 : quality \sim size$$

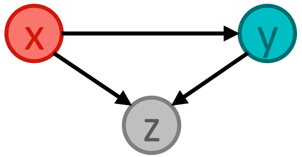
specification size

specification quality





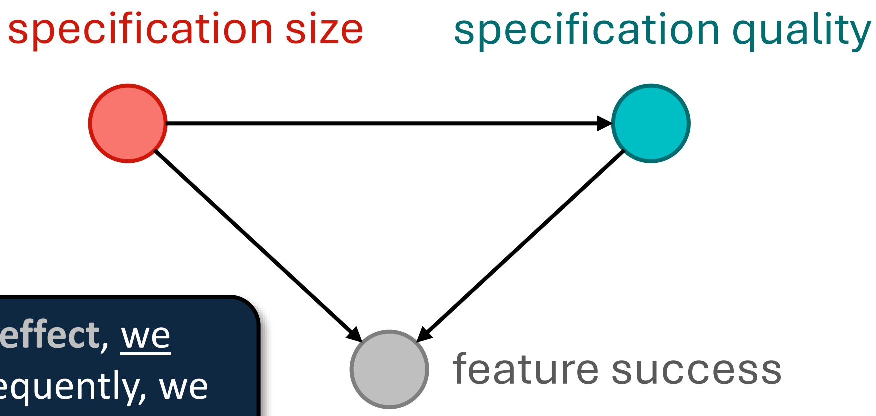
Collider



- Does the **length of a requirements specification** affect its **quality**? Both factors do affect whether the specified feature is implemented successfully.

$$\cancel{m_1: quality \sim size + success}$$

$$m_2: quality \sim size$$



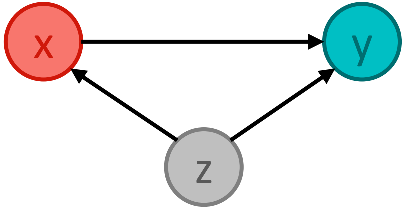
! If both **treatment** and **outcome** have a **common effect**, <u>we must not condition</u> on this common effect. Consequently, we cannot simply add all relevant variables as predictors.

The "building blocks" of causal models

Confounder

(or: fork)

z is a common cause of both exposure x and the outcome y

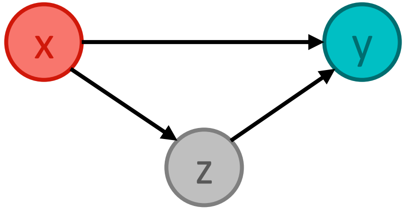


To get an unbiased estimation of the effect x → y we must control z.

Mediator

(or: pipe)

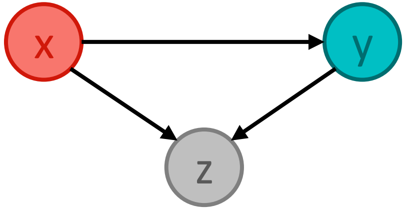
z is an effect of exposure x and a cause the outcome y



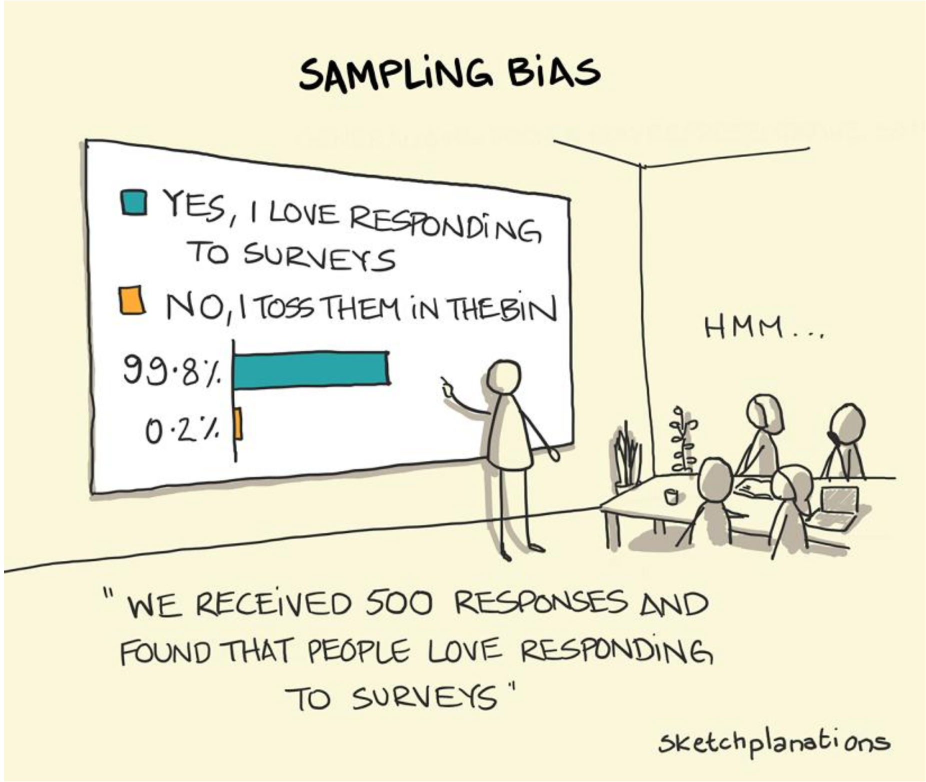
To get the direct effect x → y we must control z. To get the total effect, we must not control z.

Collider

z is a common effect of both exposure x and the outcome y



To get the direct effect x → y **DO NOT DO ANYTHING (with z)!**



https://sketchplanations.com/sampling-bias

Section 3b

d-separation

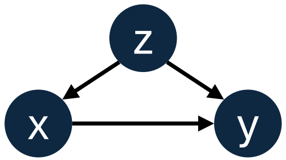
Causal Modelling

- Causal modelling is imperative for drawing the correct conclusions when conducting statistical causal inference.

! We cannot infer causality from data alone – we must also know the process that created that data.

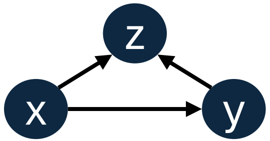
Causal Model

Statistical Model



Confounder

$$m: y \sim x + z$$



Collider

$$m: y \sim x$$

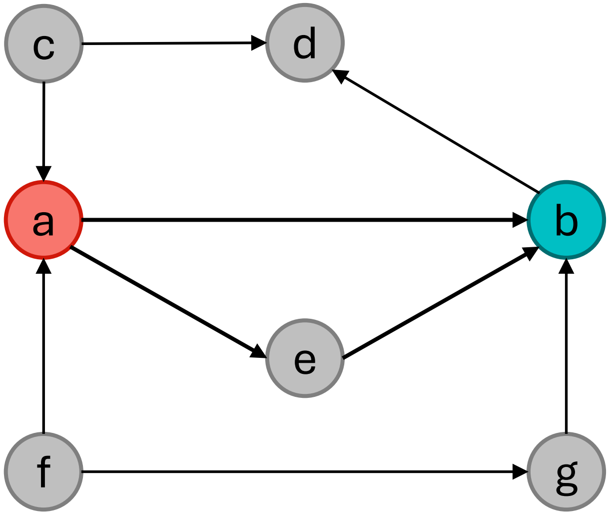
? But how does this approach scale beyond three variables?

d-separation

- Given my DAG and my **exposure** and **outcome** of interest, d-separations answers which variables to include in a statistical / ML model to get an unbiased causal estimate.
- These variables that “block confounding backdoor paths of association” are called an *adjustment set*.
- Practical intuition:
  - Include confounders (common causes of both exposure and outcome).
  - Include mediators, if you are interested in the direct effect.
  - **Do not include colliders** (common effects of both exposure and outcome).

! Once a backdoor path is closed, the flow of “anti-causal” association is blocked, and you won’t need to control for more confounding variables.

d-separation example

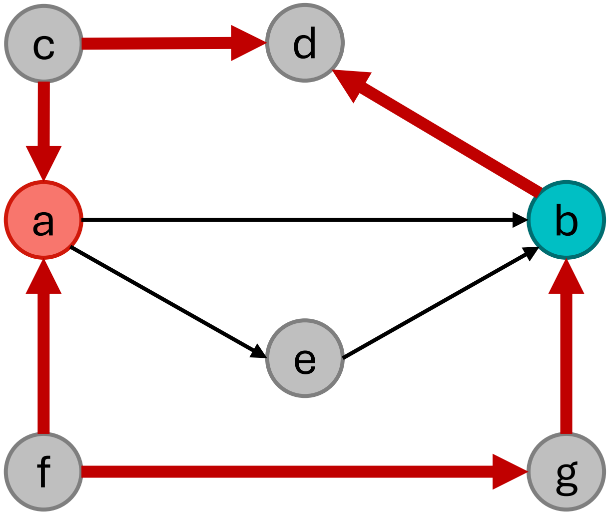


Model Derivation

$m: b \sim a$

d-separation example

Adjustment set = {}



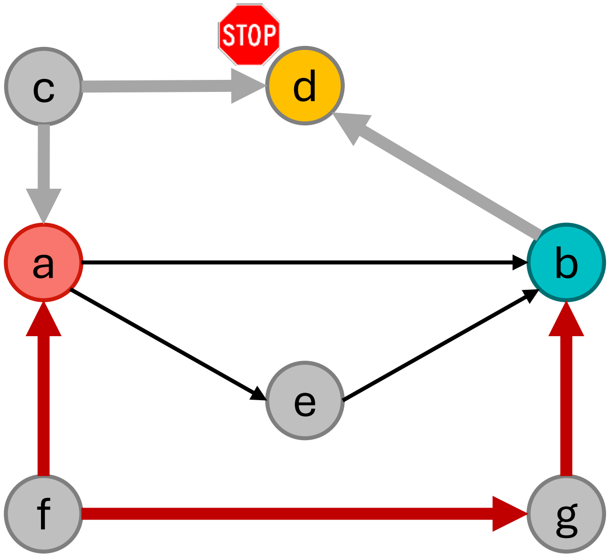
Model Derivation

$m: b \sim a$

1. Identify every backdoor path

d-separation example

Adjustment set = {}



Model Derivation

$m: b \sim a$

1. Identify every backdoor path
2. Check if there is a **collider** between the cause and effect of interest.
   a) If yes, do not condition on any of the variables on this path.

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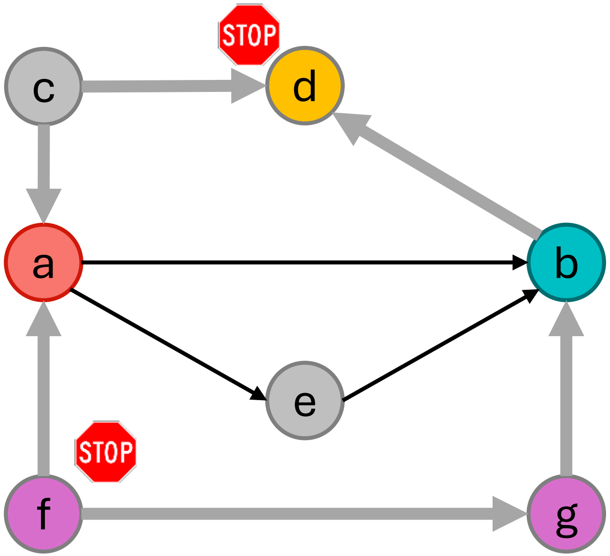
d-separation example

Adjustment set = {f}



Model Derivation

$$m\colon b \sim a + f$$

1. Identify every backdoor path
2. Check if there is a **collider** between the cause and effect of interest.
   a) If yes, do not condition on any of the variables on this path.
   b) If not, condition on **any of the variables** on this path to "close" the backdoor path.

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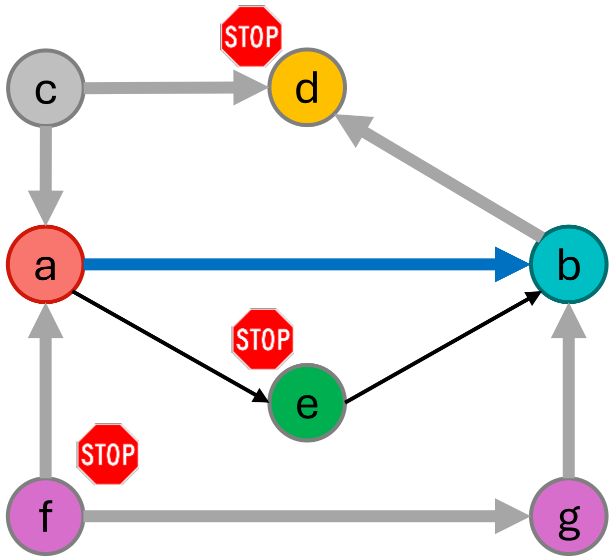
d-separation example

Adjustment set = {f,(e)}



Model Derivation

$m: b \sim a + f$

1. Identify every backdoor path
2. Check if there is a **collider** between the cause and effect of interest.
   a) If yes, do not condition on any of the variables on this path.
   b) If not, condition on **any of the variables** on this path to “close” the backdoor path.
3. Optionally, condition on **mediators** to isolate the **direct effect**.

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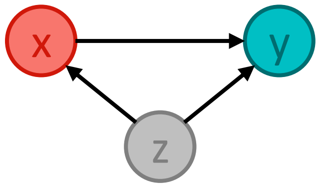
Workflow for Statistical Causal Inference

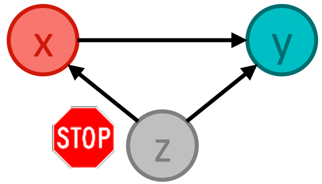
1) Modelling

Creating a DAG of assumed causal relationships



2) Identification

Define a statistical / ML model from the causal model



Model Derivation

$$y \sim x + z$$

3) Estimation

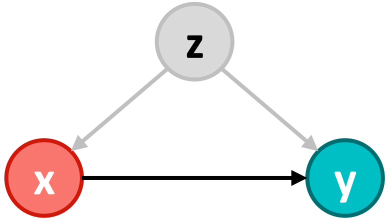
Training a statistical model with the observed data and evaluating the results

Siebert, J. (2023). Applications of statistical causal inference in software engineering. *IST*, *159*, 107198.

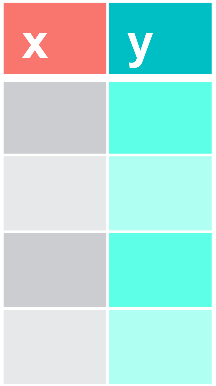
Scientific Workflow: Traditional

- Data Collection
- Modelling
- Identification
- Estimation

Assumed DAG



Collected Data



! Not collecting data about a confounder will threaten the internal validity of your results

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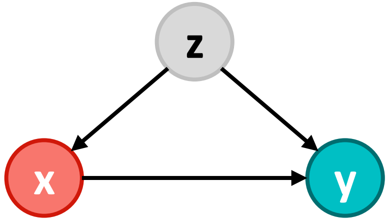
Scientific Workflow: Better

- Modelling
- Identification
- Data Collection
- Estimation

Assumed DAG

hard-to-measure confounder

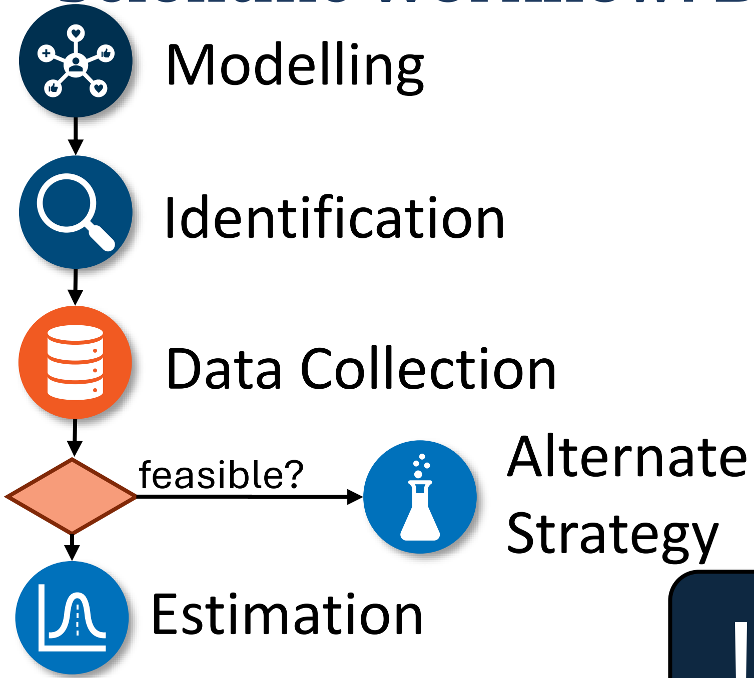


Collected Data

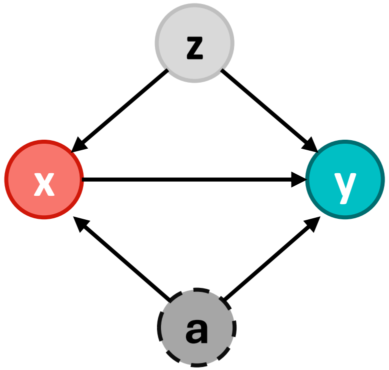
x	y	z

! Too often, we derive hypotheses from the data available to us.

Scientific Workflow: Best



Assumed DAG



Unobservable confounder

Collected Data

x	y	z	a
			X
			X
			X
			X

!

Unobservable confounders require
(a) acknowledgement in the threats to (internal) validity and (b) different research strategies.

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Summary causal structures and d-separation

- We discussed the need to model our causal assumptions about the data generating process in observational studies using DAGs.

- Confounders, Mediators, and Colliders are different structural patterns in a DAG that influence the flow of association.

- d-separation helps us to determine which variables to include or exclude in a statistical or ML model to avoid biased effect estimates.

- Causal Workflow:
  Modelling -> Identification -> Data Collection -> Estimation